Physical Optics Handbook

A Concise Reference for Fourier Optics

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Contents

1 Notation **1**
- 1.1 Coordinates . . . 1
- 1.2 Spatial Frequencies . . . 1
- 1.3 Wave Vectors . . . 1
- 1.4 Functions . . . 2
- 1.5 Operators . . . 2

2 Dirac Delta **3**
- 2.1 Definition . . . 3
- 2.2 Basic Properties . . . 3
- 2.3 Two-Dimensional Delta . . . 4
- 2.4 Delta as a Limit . . . 4
- 2.5 Delta from Complex Exponentials . . . 4

3 Fourier Transform Definitions **5**
- 3.1 Frequency Convention . . . 5
- 3.2 Wave-Vector Convention . . . 6
- 3.3 Comparison of Common Conventions . . . 6
- 3.4 Integral operation definitions . . . 6

4 Fourier Transform Properties **7**
- 4.1 Frequency convention . . . 7
  - 4.1.1 Basic Properties . . . 7
  - 4.1.2 Differential Properties . . . 8
  - 4.1.3 Integral Properties . . . 8
  - 4.1.4 Energy Relations . . . 8
- 4.2 Wave-Vector Convention . . . 9
  - 4.2.1 Basic Properties . . . 9
  - 4.2.2 Differential Properties . . . 9
  - 4.2.3 Integral Properties . . . 10
  - 4.2.4 Energy Relations . . . 10

5 Elementary Functions **11**

6 Fourier Transform Pairs **12**
- 6.1 Frequency Convention . . . 12
- 6.2 Wave-Vector Convention . . . 13

Chapter 1

Notation

1.1 Coordinates

Symbol	Meaning	Unit
(x, y, z)	Cartesian coordinates	m
(r, θ)	Polar coordinates	m, rad
(ρ, ϕ, z)	Cylindrical coordinates	m, rad
$\mathbf{r} = (x, y, z)$	Position vector	m

1.2 Spatial Frequencies

Symbol	Meaning	Unit
(f_x, f_y)	Spatial frequencies	m^{-1}
$\mathbf{f}$	Spatial frequency vector	m^{-1}

1.3 Wave Vectors

Symbol	Meaning	Unit
(k_x, k_y)	Wave-vector components	$\mathrm{rad\,m}^{-1}$
$\mathbf{k}$	Wave vector	$\mathrm{rad\,m}^{-1}$
k	Wave number	$\mathrm{rad\,m}^{-1}$
λ	Wavelength	m

The spatial frequency and wave vector are related by

$$\boxed{\mathbf{k} = 2\pi\mathbf{f}}$$

or, component-wise,

$$k_x = 2\pi f_x, \qquad k_y = 2\pi f_y.$$

1.4 Functions

Lower-case letters denote functions in the spatial domain, whereas the corresponding upper-case letters denote their Fourier transforms.

Spatial domain	Fourier domain
$u(x, y)$	$U(f_x, f_y)$
$g(x, y)$	$G(f_x, f_y)$
$h(x, y)$	$H(f_x, f_y)$
$p(x, y)$	$P(f_x, f_y)$

1.5 Operators

Symbol	Meaning
$*$	Convolution
$\star$	Cross-correlation
$\mathcal{F}\{\cdot\}$	Fourier transform
$\mathcal{F}^{-1}\{\cdot\}$	Inverse Fourier transform
∇	Gradient operator
∇^2	Laplacian operator

Chapter 2

Dirac Delta

2.1 Definition

The Dirac delta is "heuristically" defined by

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases}$$

together with the normalization

$$\int_{-\infty}^{+\infty} \delta(x)\, dx = 1.$$

Its defining property is the *sifting* property

$$\boxed{\int_{-\infty}^{+\infty} f(x)\, \delta(x - x_0)\, dx = f(x_0)}$$

which, by a change of variable, can be also interpreted as a convolution integral:

$$\boxed{\int_{-\infty}^{+\infty} f(\alpha)\, \delta(x - \alpha)\, d\alpha = f(x) * \delta(x) = f(x).}$$

Thus, the Dirac delta is the identity element of the convolution.

2.2 Basic Properties

Property	Relation
Sifting	$\int f(x)\delta(x - x_0)\, dx = f(x_0)$
Even symmetry	$\delta(-x) = \delta(x)$
Scaling	$\delta(ax) = \dfrac{1}{\lvert a \rvert}\delta(x)$
Derivative	$\int f(x)\delta'(x - x_0)\, dx = -f'(x_0)$
Product (sampling)	$f(x)\delta(x - x_0) = f(x_0)\delta(x - x_0)$

2.3 Two-Dimensional Delta

$$\delta(\mathbf{r}) = \delta(x)\delta(y)$$

with

$$\iint \delta(\mathbf{r})\, d\mathbf{r} = 1.$$

More generally,

$$\delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0)\delta(y - y_0).$$

The sifting property becomes:

$$\boxed{\iint f(x,y)\delta(x - x_0)\delta(y - y_0)\, dx\, dy = f(x_0, y_0)}$$

More generally:

$$\boxed{\iint f(\mathbf{r})\delta(\mathbf{r} - \mathbf{r}_0)\, d\mathbf{r} = f(\mathbf{r}_0)}$$

2.4 Delta as a Limit

The Dirac delta can be understood heuristically as the limit of a sequence of increasingly narrow functions with unit area.

For example,

$$\boxed{\delta(x) = \lim_{a\to\infty} a\ \mathrm{sinc}(ax)}$$

where

$$\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}.$$

Similarly, a Gaussian representation is

$$\boxed{\delta(x) = \lim_{\sigma\to 0} \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}}$$

2.5 Delta from Complex Exponentials

The Dirac delta can also be represented as

$$\boxed{\delta(x) = \int_{-\infty}^{+\infty} e^{i2\pi f x}\, df}$$

Equivalently,

$$\boxed{\delta(x) = \lim_{B\to\infty} \int_{-B}^{B} e^{i2\pi f x}\, df}$$

By solving the integral we obtain

$$\delta(x) = \lim_{B\to\infty} 2B\ \mathrm{sinc}(2Bx),$$

which is consistent with the limit discussed in the section above.

Chapter 3

Fourier Transform Definitions

3.1 Frequency Convention

One-Dimensional

Forward	$F(f) = \int_{-\infty}^{+\infty} f(x)\, e^{-j2\pi fx}\, dx$
Inverse	$f(x) = \int_{-\infty}^{+\infty} F(f)\, e^{j2\pi fx}\, df$

Two-Dimensional

Forward	$F(f_x, f_y) = \iint_{-\infty}^{+\infty} f(x,y)\, e^{-j2\pi(f_x x + f_y y)}\, dx\, dy$
Inverse	$f(x,y) = \iint_{-\infty}^{+\infty} F(f_x, f_y)\, e^{j2\pi(f_x x + f_y y)}\, df_x\, df_y$

Vector Form

$$F(\mathbf{f}) = \int f(\mathbf{r})\, e^{-j2\pi \mathbf{f}\cdot\mathbf{r}}\, d\mathbf{r}$$

$$f(\mathbf{r}) = \int F(\mathbf{f})\, e^{j2\pi \mathbf{f}\cdot\mathbf{r}}\, d\mathbf{f}$$

3.2 Wave-Vector Convention

The wave vector is related to the spatial frequency by

$$\boxed{\mathbf{k} = 2\pi\mathbf{f}}$$

One-Dimensional

Forward	$F(k) = \int_{-\infty}^{+\infty} f(x)\, e^{-jkx}\, dx$
Inverse	$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(k)\, e^{jkx}\, dk$

Two-Dimensional

Forward	$F(k_x, k_y) = \iint_{-\infty}^{+\infty} f(x, y)\, e^{-j(k_x x + k_y y)}\, dx\, dy$
Inverse	$f(x, y) = \frac{1}{(2\pi)^2} \iint_{-\infty}^{+\infty} F(k_x, k_y)\, e^{j(k_x x + k_y y)}\, dk_x\, dk_y$

3.3 Comparison of Common Conventions

Convention	Forward kernel	Inverse kernel
Frequency	$\exp(-j2\pi\mathbf{f} \cdot \mathbf{r})$	$\exp(j2\pi\mathbf{f} \cdot \mathbf{r})$
Wave vector	$\exp(-j\mathbf{k} \cdot \mathbf{r})$	$\frac{1}{(2\pi)^n} \exp(j\mathbf{k} \cdot \mathbf{r})$
Symmetric	$\frac{1}{(2\pi)^{n/2}} \exp(-j\mathbf{k} \cdot \mathbf{r})$	$\frac{1}{(2\pi)^{n/2}} \exp(j\mathbf{k} \cdot \mathbf{r})$

3.4 Integral operation definitions

Convolution	$(f * g)(x, y) = \iint f(\xi, \eta)\, g(x - \xi, y - \eta)\, d\xi\, d\eta$
Cross-correlation	$(f \star g)(x, y) = \iint f^*(\xi, \eta)\, g(\xi + x, \eta + y)\, d\xi\, d\eta$

Chapter 4

Fourier Transform Properties

4.1 Frequency convention

Throughout this chapter,

$$f(x,y) \overset{\mathcal{F}}{\longleftrightarrow} F(f_x, f_y).$$

4.1.1 Basic Properties

Property	Spatial Domain	Fourier Domain
Linearity	$a\,f + b\,g$	$aF + bG$
Complex Conjugation	$f^*(x,y)$	$F^*(-f_x,-f_y)$
Duality	$F(x,y)$	$f(-f_x,-f_y)$
Hermitian	$f(x,y) \in \Re$	$F(-f_x,-f_y) = F(f_x,f_y)$
Shift	$f(x-x_0, y-y_0)$	$F(f_x,f_y)e^{-j2\pi(f_x x_0 + f_y y_0)}$
Modulation	$f(x,y)e^{j2\pi(f_{0x}x + f_{0y}y)}$	$F(f_x - f_{0x}, f_y - f_{0y})$
Stretch	$f(ax,by)$	$\frac{1}{\lvert ab\rvert} F\left(\frac{f_x}{a}, \frac{f_y}{b}\right)$
Rotation matrix R	$f(R\mathbf{r})$	$F(R\mathbf{f})$
Separability	$f(x,y) = f_x(x)f_y(y)$	$F_x(f_x)F_y(f_y)$

4.1.2 Differential Properties

Property	Spatial Domain	Fourier Domain
$\partial/\partial x$	$\dfrac{\partial f}{\partial x}$	$j2\pi f_x F$
$\partial/\partial y$	$\dfrac{\partial f}{\partial y}$	$j2\pi f_y F$
Multiplication by x	$x\, f(x,y)$	$\dfrac{j}{2\pi}\dfrac{\partial F}{\partial f_x}$
Multiplication by y	$y\, f(x,y)$	$\dfrac{j}{2\pi}\dfrac{\partial F}{\partial f_y}$
Gradient	∇f	$j2\pi \mathbf{f}\, F$
Laplacian	$\nabla^2 f$	$-4\pi^2 \lvert\mathbf{f}\rvert^2 F$

4.1.3 Integral Properties

Property	Spatial Domain	Fourier Domain
Convolution	$f * g$	FG
Multiplication	fg	$F * G$
Cross-correlation	$f \star g$	FG^*
Integral	$\iint f(x,y)\, dx\, dy$	$F(0,0)$

4.1.4 Energy Relations

Property	Expression
Parseval	$\iint \lvert f(x,y)\rvert^2 dxdy = \iint \lvert F(f_x,f_y)\rvert^2 df_x df_y$
Plancherel	$\iint f(x,y)g^*(x,y)\, dxdy = \iint F(f_x,f_y)G^*(f_x,f_y)\, df_x df_y$

4.2 Wave-Vector Convention

Throughout this section,

$$f(x,y) \overset{\mathcal{F}}{\longleftrightarrow} F(k_x,k_y),$$

with

$$\mathbf{k} = 2\pi\mathbf{f}.$$

4.2.1 Basic Properties

Property	Direct Domain	Fourier Domain
Linearity	$a\,f + b\,g$	$aF + bG$
Complex Conjugation	$f^*(x,y)$	$F^*(-k_x,-k_y)$
Duality	$F(x,y)$	$(2\pi)^2\,f(-k_x,-k_y)$
Hermitian	$f(x,y) \in \Re$	$F(-f_x,-f_y) = F(f_x,f_y)$
Shift	$f(x-x_0,y-y_0)$	$F(k_x,k_y)e^{-j(k_x x_0+k_y y_0)}$
Modulation	$f(x,y)e^{j(k_{0x}x+k_{0y}y)}$	$F(k_x-k_{0x},k_y-k_{0y})$
Stretch	$f(ax,by)$	$\frac{1}{\lvert ab\rvert}F\left(\frac{k_x}{a},\frac{k_y}{b}\right)$
Rotation matrix R	$f(R\mathbf{r})$	$F(R\mathbf{k})$
Separability	$f(x,y) = f_x(x)f_y(y)$	$F_x(f_x)F_y(f_y)$

4.2.2 Differential Properties

Property	Direct Domain	Fourier Domain
$\partial/\partial x$	$\frac{\partial f}{\partial x}$	jk_xF
$\partial/\partial y$	$\frac{\partial f}{\partial y}$	jk_yF
Multiplication by x	$x\,f$	$j\frac{\partial F}{\partial k_x}$
Multiplication by y	$y\,f$	$j\frac{\partial F}{\partial k_y}$
Gradient	∇f	$j\mathbf{k}\,F$
Laplacian	$\nabla^2 f$	$-\lvert\mathbf{k}\rvert^2F$

4.2.3 Integral Properties

Property	Direct Domain	Fourier Domain
Convolution	$f * g$	FG
Multiplication	fg	$\frac{1}{(2\pi)^2}(F * G)$
Cross-correlation	$f \star g$	FG^*
Integral	$\iint f(x,y)\,dx\,dy$	$F(0,0)$

4.2.4 Energy Relations

Property	Expression
Parseval	$\iint \lvert f\rvert^2\,dx\,dy = \frac{1}{(2\pi)^2}\iint \lvert F\rvert^2\,dk_x\,dk_y$
Plancherel	$\iint f\,g^*\,dx\,dy = \frac{1}{(2\pi)^2}\iint FG^*\,dk_x\,dk_y$

Chapter 5

Elementary Functions

This chapter summarizes the elementary functions most frequently used in Fourier optics.

Function	Definition
Rectangle	$\mathrm{rect}(x) = \begin{cases} 1, & \lvert x\rvert < \frac{1}{2} \\ \frac{1}{2}, & \lvert x\rvert = \frac{1}{2} \\ 0, & \lvert x\rvert > \frac{1}{2} \end{cases}$
Triangle	$\mathrm{tri}(x) = \begin{cases} 1 - \lvert x\rvert, & \lvert x\rvert < 1 \\ 0, & \lvert x\rvert \geq 1 \end{cases}$
Dirac delta	$\delta(x), \int \delta(x) f(x) dx = f(0)$
Dirac comb	$\mathrm{comb}(x) = \sum_{n=-\infty}^{+\infty} \delta(x - n)$
Unit step	$u(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}, & x = 0 \\ 1, & x > 0 \end{cases}$
Sign	$\mathrm{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$
Sinc	$\mathrm{sinc}(x) = \dfrac{\sin(\pi x)}{\pi x}$
Gaussian	$g(x) = e^{-\pi x^2}$
Cirle	$\mathrm{circ}(r) = \begin{cases} 1, & r < 1 \\ \frac{1}{2}, & r = 1 \\ 0, & r > 1 \end{cases}$
Jinc (or sombrero)	$\mathrm{jinc}(r) = 2\dfrac{J_1(\pi r)}{\pi r}$
Complex exponential	$e^{i2\pi f_0 x}$

Chapter 6

Fourier Transform Pairs

Throughout this chapter,

$$f(x,y) \overset{\mathcal{F}}{\longleftrightarrow} F(f_x, f_y).$$

6.1 Frequency Convention

Function $f(x,y)$	Fourier Transform $F(f_x, f_y)$	Notes
1	$\delta(f_x)\delta(f_y)$	Constant
$\delta(x)\delta(y)$	1	Dirac delta
$\operatorname{rect}\left(\frac{x}{a}\right)\operatorname{rect}\left(\frac{y}{b}\right)$	$\lvert ab\rvert\ \operatorname{sinc}(af_x)\operatorname{sinc}(bf_y)$	Rectangular aperture
$\operatorname{tri}\left(\frac{x}{a}\right)\operatorname{tri}\left(\frac{x}{b}\right)$	$\lvert ab\rvert \operatorname{sinc}^2(af_x)\operatorname{sinc}^2(bf_y)$	Triangle
$e^{-\pi(x^2/a^2+y^2/b^2)}$	$\lvert ab\rvert\, e^{-\pi(a^2f_x^2+b^2f_y^2)}$	Gaussian
$\operatorname{comb}(ax)\operatorname{comb}(by)$	$\frac{1}{\lvert ab\rvert}\operatorname{comb}\left(\frac{f_x}{a}\right)\operatorname{comb}\left(\frac{f_y}{b}\right)$	Dirac comb
$e^{j2\pi(f_{0x}x+f_{0y}y)}$	$\delta(f_x-f_{0x})\delta(f_y-f_{0y})$	Plane wave
$\operatorname{circ}\left(\frac{r}{R}\right)$	$2\pi R^2\frac{J_1(2\pi R\rho)}{2\pi R\rho} = \pi R^2 \operatorname{jinc}(2R\rho)$	Circular aperture

where

$$r = \sqrt{x^2+y^2}, \qquad \rho = \sqrt{f_x^2+f_y^2}.$$

6.2 Wave-Vector Convention

Using

$$\mathbf{k} = 2\pi\mathbf{f},$$

the corresponding transform pairs are

Function $f(x,y)$	Fourier Transform $F(k_x,k_y)$	Notes
1	$(2\pi)^2\delta(k_x)\delta(k_y)$	Constant
$\delta(x)\delta(y)$	1	Dirac delta
$\mathrm{rect}\left(\frac{x}{a}\right)\mathrm{rect}\left(\frac{y}{b}\right)$	$\lvert ab\rvert\ \mathrm{sinc}\left(\frac{ak_x}{2\pi}\right)\mathrm{sinc}\left(\frac{bk_y}{2\pi}\right)$	Rectangular aperture
$\mathrm{tri}\left(\frac{x}{a}\right)\mathrm{tri}\left(\frac{x}{b}\right)$	$\lvert ab\rvert\,\mathrm{sinc}^2\left(\frac{ak_x}{2\pi}\right)\mathrm{sinc}^2\left(\frac{bk_y}{2\pi}\right)$	Triangle
$e^{-\pi(x^2/a^2+y^2/b^2)}$	$\lvert ab\rvert\, e^{-\frac{1}{4\pi}(a^2k_x^2+b^2k_y^2)}$	Gaussian
$\mathrm{comb}(ax)\,\mathrm{comb}(by)$	$\frac{1}{\lvert ab\rvert}\mathrm{comb}\left(\frac{k_x}{2\pi a}\right)\mathrm{comb}\left(\frac{k_y}{2\pi b}\right)$	Dirac comb
$e^{j(k_{0x}x+k_{0y}y)}$	$(2\pi)^2\delta(k_x-k_{0x})\delta(k_y-k_{0y})$	Plane wave
$\mathrm{circ}\left(\frac{r}{R}\right)$	$2\pi R^2\frac{J_1(Rk)}{Rk} = \pi R^2\,\mathrm{jinc}\left(\frac{Rk}{\pi}\right)$	Circular aperture

where

$$r = \sqrt{x^2+y^2}, \qquad k = \sqrt{k_x^2+k_y^2}.$$

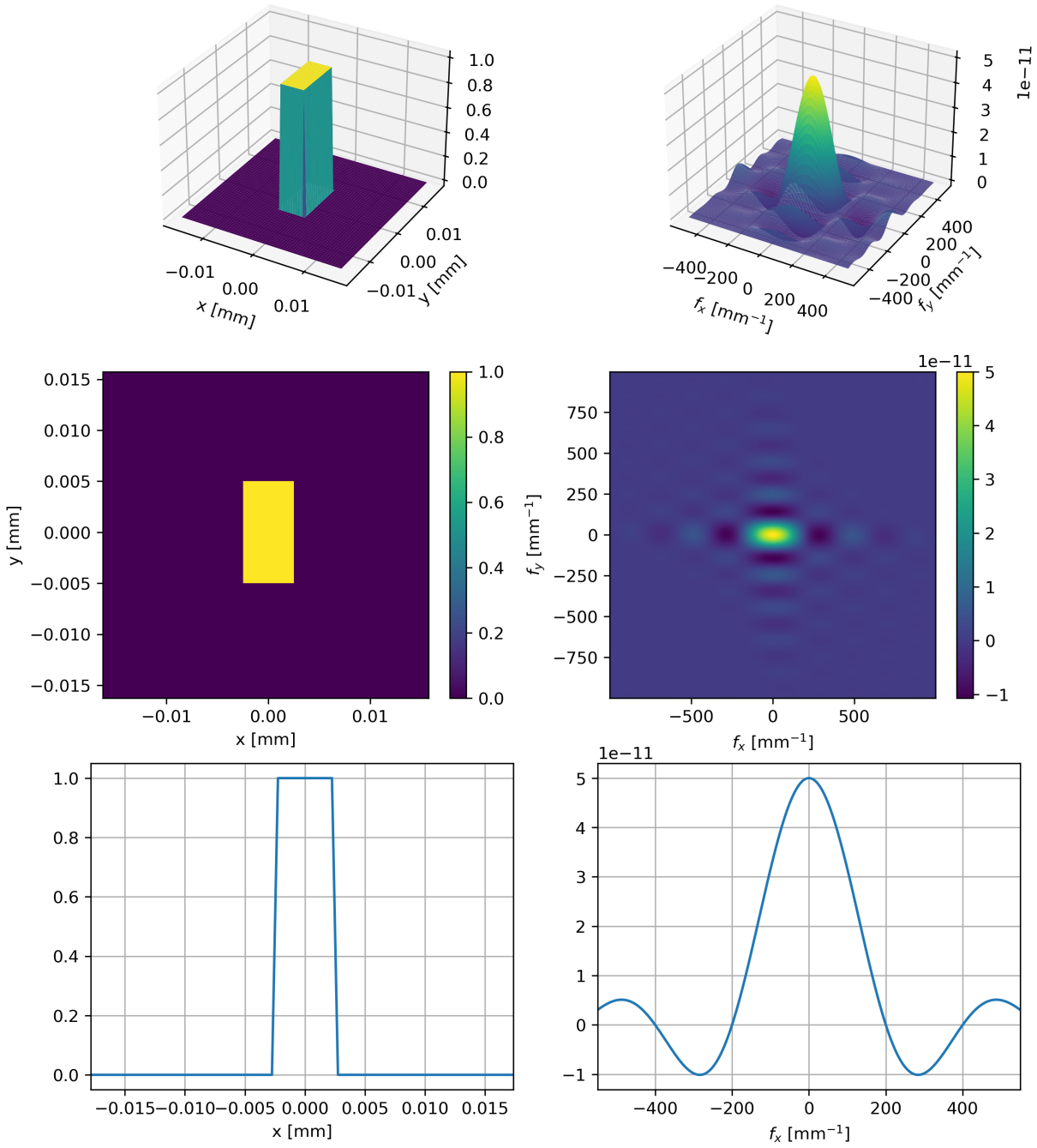


Figure 6.1: Fourier Transform of a rectangle function with $a = 5\mu$m and $b = 10\mu$m. On the left column the rectangle, on the right column the Fourier transform.

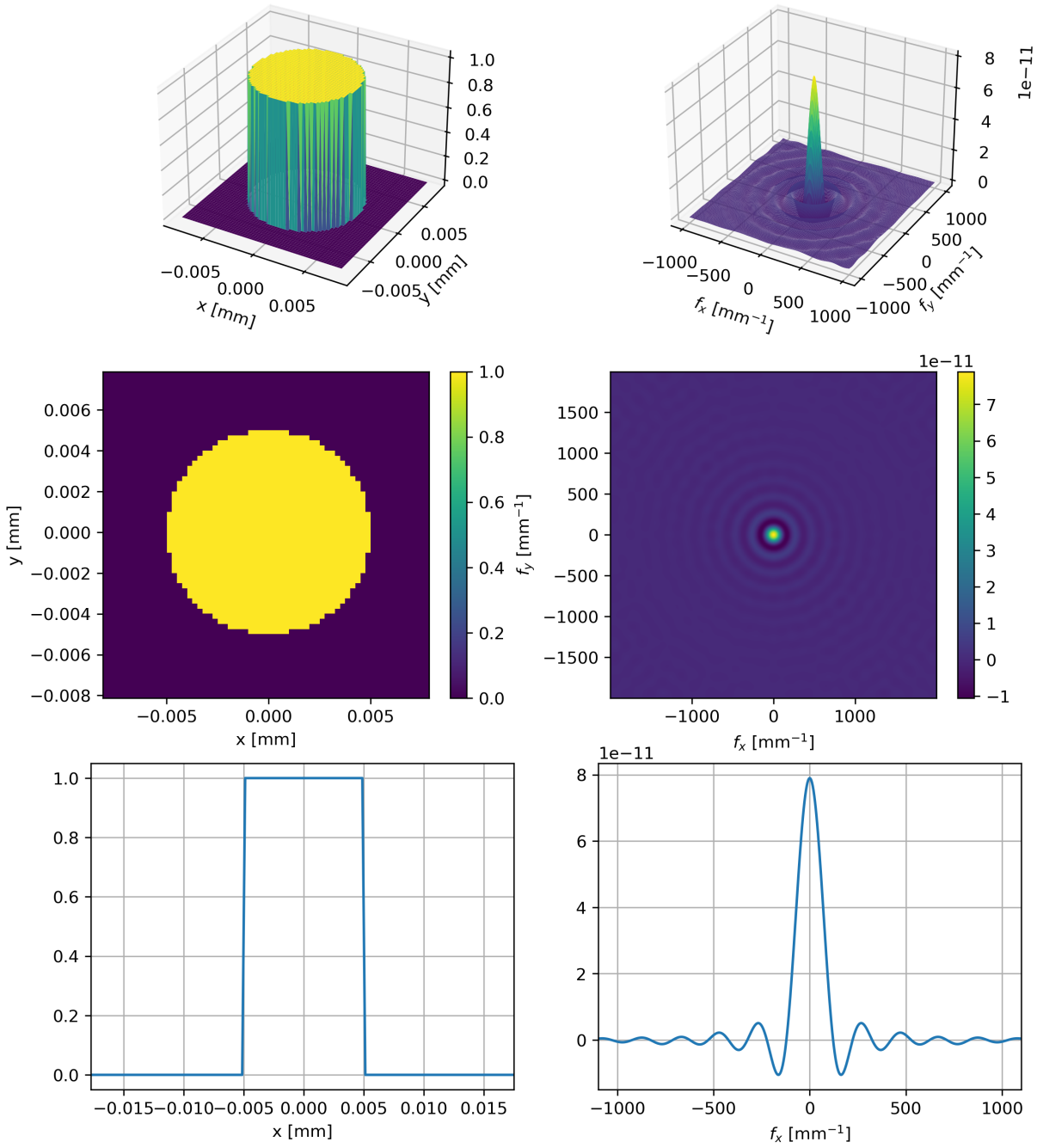


Figure 6.2: Fourier Transform of a circle function with $R = 5\mu$m. On the left column the circle, on the right column the Fourier transform.